

Causal Inference 2026
Assignment 1
Analysis of Classical Randomized Experiments

Due: Friday, April 24th 2026

Guidelines

This assignment requires the use of R. Do not include code in your written report: instead, provide a clear and concise presentation of your results, avoiding copy-pasted R output.

The report must not exceed **5000 characters (including spaces)**, excluding tables and figures. All tables and figures must be numbered and placed at the end of the report.

The report may be written in Italian or English and must be submitted in PDF format. Partial credit based on your code may be awarded for some questions.

- Upload both your report and R code file on Moodle before the deadline.
- Use the following file naming convention:

ASS1_GroupX.pdf and ASS1RCode_GroupX.R,

where X is the group number (see the google file)

- Include the name and surname of all group members in both the PDF report and the R script.

Data Description

The dataset `Morphine.txt` (available on Moodle) contains data on a randomized experiment, involving adult patients who were undergoing an elective open colorectal abdominal surgery.

Patients were randomized to receive, before surgery, the experimental treatment or an active placebo and the outcome of primary interest was postoperative pain intensity, measured using a visual analogue scale (VAS). According to the medical guidelines for pain control, after surgery patients received an Intra-Venous Patient Controlled Analgesia (IV-PCA) system programmed to give off fixed doses of morphine sulphate upon patient demand.

The dataset includes the following variables:

- Covariate: **sex** - patient's gender (0=female, 1=male); **age** - patient's age
- Treatment variable: **W** (0=placebo, 1=new drug);

- Outcome variables (measured at 4 hours from the end of surgery)
 - AS4: number of self-administered postoperative doses of morphine sulphate
 - VS4: Static VAS (VAS scores at rest)
 - VD4: Dynamic VAS (VAS scores on movement, that is, upon coughing)

Consider **VD4 as outcome variable**.

Questions

Answer questions 1-4 assuming that the data come from a completely randomized experiment.

1. **Descriptive analysis:** Briefly comment the results.
 - (a) Summarize the data with descriptive statistics (including summary statistics and graphs)
 - (b) Inspect the dataset, and provide some evidence that the assignment was indeed random, e.g. by examining the pre-treatment covariate distributions.
2. **Fisher:** Briefly comment the results.

- (a) Calculate exact (two sided) Fisher p-value for the sharp null hypothesis of no effects whatsoever using the difference in mean statistic. Approximate the p-value using a Monte Carlo approximation. Then calculate the asymptotic p-value.
- (b) Calculate a 90% Fisher interval for a constant additive treatment effect using the difference in mean statistic.
- (c) Calculate exact (two sided) Fisher p-value for the sharp null hypothesis of no effects whatsoever using a statistic that adjusts for covariates, through either a model output strategy or a pseudo-outcome strategy (not both)
- (d) Calculate a 90% Fisher interval for a constant additive treatment effect using a statistic that adjusts for covariates, through either a model output strategy or a pseudo-outcome strategy (not both). Use the same strategy you have chosen for point 2c.

Hint: If the null hypothesis is that the treatment effect is $H_0 : Y_i(1) - Y_i(0) = \tau$, then we first calculate for each unit the implied value of $Y_i(0)$. For units with $W_i = 0$, we have $Y_i(0) = Y_i^{obs}$, and for units $W_i = 1$, we have $Y_i(0) = Y_i^{obs} - \tau$ under the null hypothesis. Then use these $Y_i(0)$ as dependent variable in implementing the model output strategy or the pseudo-outcome strategy

3. **Neyman and regression:** Briefly comment the results.
 - (a) Calculate an unbiased estimate of the average treatment effect.
 - (b) Apply Neyman's method to construct a 90% large sample confidence interval for the average treatment effect.

- (c) Use Neyman's hypothesis testing to test $H_0 : \bar{Y}(1) - \bar{Y}(0) = 0$ versus $H_1 : \bar{Y}(1) - \bar{Y}(0) \neq 0$.
- (d) Use Lin's ANCOVA estimator to get an estimate of the average treatment effect and its variance.

4. **Bayesian model-based inference.** Assume that

$$\begin{bmatrix} Y_i(0) \\ Y_i(1) \end{bmatrix} \mid \mathbf{X}_i, \theta \sim \mathcal{N} \left(\begin{bmatrix} \beta_c + \beta_c^{(X)'} \mathbf{X}_i \\ \beta_t + \beta_t^{(X)'} \mathbf{X}_i \end{bmatrix}, \begin{bmatrix} \sigma_c^2 & 0 \\ 0 & \sigma_t^2 \end{bmatrix} \right)$$

where $\mathbf{X}_i = (\text{Gender}, \text{Age})$

- (a) Assume parameter are a priori independent and specify weakly informative priors (normal priors for the regression coefficients and scaled $\text{inv-}\chi^2$ priors for the variances). Details on the selected priors must be included in the report. Construct a Gibbs Sampler to make inference on the mixed-population and finite-sample average treatment effect for the morphine data. Run it for 15 000 iterations. Derive the posterior mean, variance, and 90% credible interval for the mixed and finite-sample average treatment effect. Plot the posterior distributions of the causal estimands. Briefly comment the results.

Answer questions 5-7 assuming that the data come from a stratified randomized experiment with strata defined by gender.

- 5. **Fisher.** Calculate exact (two sided) Fisher p-value for the sharp null hypothesis of no effects whatsoever using as test statistic the weighted average of the strata-specific differences-in-means with weights given by the relative sample size in the strata. Approximate the p-value using a Monte Carlo approximation.
- 6. **Neyman.** Calculate an unbiased estimate of the average treatment effect and a conservative estimate of the variance of the estimator
- 7. **Regression.** Use Lin's ANCOVA estimator for SRE to get an estimate of the average treatment effect and its variance.